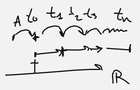
$(a_n)_n$, $\sum_{n=1}^{\infty} a_n = \lim_{N\to+\infty} \sum_{n=1}^{N} a_n$

$b_n = \sum_{k=1}^{n} a_k$, (b_n) cv ⇔



$\sum t_n < +\infty$

$b_n = \sum_{k=2}^{n} a_k \to +\infty$ $\quad b_n = (1)^n$

$a_n = b_n - b_{n-1}$

$\sum_{k=1}^{n} 1 \to n \to +\infty$ $\quad b_n = \sum_{k=1}^{n} a_k + a_1$

a- $\sum_{k=0}^{+\infty} (1/3)^n$ $\quad$ b- $\sum_{n=1}^{\infty} (1/\sqrt{3})^{n+3}$ $\quad$ c- $\sum_{n=1}^{\infty} 5^n$

$\lim_N \sum_{n=0}^{N} (1/3)^n = \lim_{N\to+\infty} \frac{1-(1/3)^{N+1}}{1-(1/3)} = \frac{1-0}{1-1/3}$

$= 1 \Big/ \frac{3}{3} - \frac{1}{3} = 1 \Big/ \frac{2}{3} = 3/2.$

$\sum_{k=1}^{N+1} (1/3)^n - \sum_{k=0}^{N} (1/3)^n = (1/3)^{N+1} - 1$ ⟷ $\left\{ \sum_{n=0}^{N} (1/3)^n = \frac{1-(1/3)^{N+1}}{1-(1/3)} \right.$

$= \sum_{n=0}^{N} (1/3)^{n+1} - \sum_{k=0}^{N} (1/3)^n = \sum_{n=0}^{N} (1/3)^{n+1} - (1/3)^n = \sum_{n=0}^{N} (1/3)^n \underbrace{[1/3 - 1]}_{-2/3} = (1/3 - 1) \sum_{n=0}^{N} (1/3)^n$

$\sum_{n=1}^{N} (\lambda)^{n+3} = \sum_{n=0}^{N-1} \lambda^{n+4} = \sum_{k=0}^{N-1} \lambda^4 \lambda^n = \lambda^4 \underbrace{\sum_{n=0}^{N-1} \lambda^n}_{} = \lambda^4 \left(\frac{1-\lambda^N}{1-\lambda} \right) \to \frac{\lambda^4}{1-\lambda}$

$\lambda = (1/\sqrt{3})$ $\quad \sqrt{3} > 1$, $1/\sqrt{3} < 1$

$= \frac{(1/\sqrt{3})^4}{1-(1/\sqrt{3})} = \frac{(1/3)^2}{1-(1/\sqrt{3})} = \frac{1/9}{1-1/\sqrt{3}}$

$= \frac{1}{9(1-1/\sqrt{3})} = \frac{1}{9-3\sqrt{3}}$

$9 = 3^2 = (\sqrt{3}^2)^2 = \sqrt{3}^4$

$(\sqrt{x})^2 = x$

c- $\sum_{n=1}^{\infty} 5^{n+2}$, $5^n \to +\infty$

$\sum_{n=0}^{\infty} a_n$ $\quad a_n \to k \neq 0$

$a_n \to k$, $\forall n \geq n_0$, $k - \varepsilon \leq a_n \leq k + \varepsilon$

$\sum_{0}^{N} a_n = \underbrace{\sum_{0}^{n_0} a_n}_{<+\infty} + \sum_{n_0+1}^{N} \underbrace{a_n}_{\geq k-\varepsilon > 0} \geq \sum_{n_0+1}^{N} (k-\varepsilon) = \underbrace{(N - (n_0+1))}_{\to+\infty} \underbrace{(k-\varepsilon)}_{>0} \to +\infty$

$a_n \to k \neq 0 \Rightarrow \sum a_n$ DV

$a_n \to 0 \Leftarrow \sum a_n$ a